Data Analytics Internship Weekly Assignment Report

Week-5 Task

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Domain : Data Analytics

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Task1: Power BI Introduction & Data Import

1. Introduction to Power BI

Power Bi is a Business Intelligence and Data visualization toll developed by Microsoft .
It is Used to:

- Import Data From different sources .
- Clean and transform data .
- Create relationship between tables .
- Perform calculation using DAX .
- Create interactive Dashboard and reports .
- Analyze data and find business insights .
- Share reports with other .

Basic purpose of Power BI in Data Analytics

The main purpose of Power BI is to convert raw data into a meaningful insights using interactive reports and visualization .

The typical Power BI workflow is :
Data Source → Data Import → Data Cleaning → Data Modeling → Analysis → Visualization → Dashboard/Report

2. Exploring the Power BI Interface

When we open Power BI Desktop, we will see several important sections .
Report View
Used to create:

- Charts
- Tables
- Cards
- KPIs
- Slicers
- Dashboards and interactive reports

Data View
Used to:

- View imported data.
- Check rows and columns.
- Inspect data values.
- Understand the dataset structure.

Model View
Used to:

- Create relationships between multiple tables.
- View the data model.
- Manage table connections.

Power Query Editor
Used for:
- Cleaning data.
- Removing duplicates.
- Handling missing values.
- Changing data types.
- Renaming columns.
- Creating new columns.

3. Importing a Dataset into Power BI
For this task, you can use a suitable dataset such as:
- Superstore Sales Dataset
- Retail Sales Dataset
- Marketing Campaign Dataset
- Healthcare Dataset

Steps to Import Data
1. Open **Power BI Desktop**.
2. Click **Get Data**.
3. Select the appropriate data source, such as **Excel** or **Text/CSV**.
4. Browse and select your dataset.
5. Select the required worksheet or table.
6. Click **Load** to directly import the data.

If data cleaning is required before importing, click **Transform Data**.

4. Viewing the Imported Data
After importing the dataset:
1. Click the **Data View** icon from the left panel.
2. Select the imported table.
3. View all available:
   - Rows
   - Columns
   - Values

This helps us understand the structure and content of the dataset.

5. Identifying Available Tables and Columns
The imported tables are visible in the **Data Pane** or **Fields Pane**.

6. Checking Data Types
It is important to verify that every column has the correct data type.

How to Check Data Types
We can check data types in:
Power Query Editor → Select Column → Data Type
or
Data View → Column Tools → Data Type

Correct data types are important because incorrect data types can cause problems in calculations, sorting, filtering, and visualizations.

Task 2: Data Cleaning & Preparation

Data cleaning is the process of identifying and correcting problems in raw data before analysis.
In Power BI, data cleaning is mainly performed using **Power Query Editor**.

What is Power Query Editor ?

Power Query is a data transformation and **ETL** tool in Power BI.
It is used to:

- Remove duplicates
- Handle missing values
- Change data types
- Merge tables
- Append tables
- Split columns
- Pivot and unpivot data

Power Query uses the **M** language behind the scenes.

What is ETL ?
ETL = Extract, Transform, Load

- **Extract:** Get data from different sources.
- **Transform:** Clean and modify the data.
- **Load:** Load the prepared data into the data model.

Example:
Extract data from Excel → Transform it in Power Query → Load it into Power BI

1. Open Power Query Editor

1. In Power BI Desktop, click Transform Data.
2. Power Query Editor will open.

3. Perform all necessary cleaning and transformation steps.

2. Check for Missing or Null Values

Identify columns containing:

- Null values
- Blank values
- Missing records

Actions

Depending on the column:

- Remove rows with important missing values.
- Replace null values with appropriate values.
- Fill missing categorical values where appropriate.

For example:

- Missing Sales → investigate or remove if invalid.
- Missing Customer Name → replace only if a valid business rule exists.
- Missing numerical values → replace with 0 only when logically appropriate.

3. Remove Duplicate Records

Duplicate records can produce incorrect results during analysis.

Steps

1. Select the required columns or entire table.
2. Go to **Home**.
3. Click **Remove Rows**.
4. Select **Remove Duplicates**.

This ensures that repeated records are removed

4. Correct Data Types

Check every column and assign the appropriate data type.

Examples:

- Dates → Date
- Quantity → Whole Number
- Sales → Decimal Number
- Profit → Decimal Number
- Names and Categories → Text

This is important for accurate calculations and analysis.

5. Rename Columns

Rename unclear or poorly formatted column names.

For example:

- Cust_Name → Customer Name
- Ord_Date → Order Date

- Sales_Amt → Sales Amount

Clear column names improve report readability.

6. Remove Unnecessary Columns

Columns that are not required for analysis should be removed.

Steps

1. Select unnecessary columns.
2. Right-click.
3. Select **Remove Columns**.

This makes the dataset easier to manage and can improve performance.

7. Handle Text Formatting Issues

Clean text columns by:

- Removing extra spaces.
- Correcting capitalization.
- Replacing incorrect values.
- Standardizing category names.

Example

Before cleaning:

- Furniture
- furniture
- FURNITURE

After cleaning:

- Furniture

Power Query options:

Transform → Format → Trim

and

Transform → Format → Clean

8. Check for Errors

Power Query can identify data errors.

You should:

1. Check each important column.
2. Identify error values.
3. Correct or remove invalid records.

This improves data quality before visualization.

Task 3: Data Modeling & Relationships

Objective
The objective of this task is to create relationships between multiple tables using common columns and build a proper **data model** in Power BI.

What is Data Modeling in Power BI?
Data Modeling is the process of organizing data from different tables and connecting those tables in a meaningful way so that Power BI can analyze the data correctly.
In simple words:
Data Modeling = Organizing tables + Creating relationships between them.

What is a Relationship?
A **relationship** is a connection between two tables using a common column.

Types of Relationships in Power BI
1. One-to-One (1:1)
One record in Table A is connected to only one record in Table B.

2. One-to-Many (1:*)
One record in one table is connected to multiple records in another table.

3. Many-to-One (*:1)
This is the same relationship viewed from the opposite direction.

4. Many-to-Many (*:*)
Multiple records in both tables can match multiple records in the other table.

How to Create the Relationship in Power BI
Method 1: Model View

1. Click **Model View** from the left panel.
2. Identify the common column between two tables.
3. Drag the common column from one table to the corresponding column in another table.
4. Power BI opens the relationship window.
5. Select the correct cardinality:

Many to One (*:1)

6. Select the appropriate cross-filter direction.
7. Click **OK**.

Method 2: Manage Relationships

1. Go to the **Modeling** tab.
2. Click **Manage Relationships**.
3. Click **New Relationship**.
4. Select the first table and column.
5. Select the related table and common column.
6. Choose:

Cardinality: Many to One (*:1)

7. Set the relationship as **Active**.
8. Click **Save/OK**.

Explanation of the Created Data Model

I have use Medical Table .

The data model follows a structure where:

Fact Tables

These tables contain transactional or repeated records:

- Diagnostics_Fact
- Pharmacy_Fact
- Visits_Fact

Dimension/Master Tables

These contain unique descriptive information:

- Patient_Registry
- Medical_Staff
- Department_Master

The fact tables are connected to the dimension tables using common keys such as:

- Patient Code
- Staff Code
- Department Code/Name

This structure helps Power BI perform accurate:

- Filtering
- Aggregation
- DAX calculations
- Data analysis
- Interactive reporting

1. Identify Common Columns Between Tables

The following common columns are used to connect the tables:

Fact Table	Common Column	Related Table	Common Column
Diagnostics_Fact	Patient Ref	Patient_Registry	Patient_Code
Medical_Staff	Department_Name	Department_Master	Department column
Pharmacy_Fact	Patient_Ref	Patient_Registry	Patient_Code
Pharmacy_Fact	Prescribed_By	Medical_Staff	Staff_Code
Visits_Fact	Consultant_Code	Medical_Staff	Staff_Code
Visits_Fact	Registered_Patient	Patient_Registry	Patient_Code

2. Relationships Created

Your current data model contains the following relationships:

Relationship 1:

Diagnostics Fact → Patient Registry

Diagnostics_Fact[Patient_Reference]

Many (*)

↓

One (1)

Patient_Registry[Patient_Code]

Purpose:

One patient can have multiple diagnostic records, but each diagnostic record belongs to one patient.

Relationship Type: Many-to-One (*:1)

Relationship 2:

Medical Staff → Department Master

Medical_Staff[Department_Name]

Many (*)

↓

One (1)

Department_Master[Department]

Purpose:

One department can have multiple medical staff members, but each staff member belongs to one department.

Relationship Type: Many-to-One (*:1)

Relationship 3:

Pharmacy Fact → Patient Registry

Pharmacy_Fact[Patient_Ref]

Many (*)

↓

One (1)

Patient_Registry[Patient_Code]

Purpose:

A patient may receive multiple prescriptions or pharmacy transactions.

Relationship Type: Many-to-One (*:1)

Relationship 4:

Pharmacy Fact → Medical Staff

Pharmacy_Fact[Prescribed_By]

Many (*)

↓

One (1)

Medical_Staff[Staff_Code]

Purpose:

One doctor or medical staff member can prescribe medicines to multiple patients.

Relationship Type: Many-to-One (*:1)

Relationship 5:

Visits Fact → Medical Staff

Visits_Fact[Consultant_Code]

Many (*)

↓

One (1)

Medical_Staff[Staff_Code]

Purpose:

One medical staff member or consultant can attend multiple patient visits.

Relationship Type: Many-to-One (*:1)

Relationship 6:

Visits Fact → Patient Registry

Visits_Fact[Registered_Patient]

Many (*)

↓

One (1)

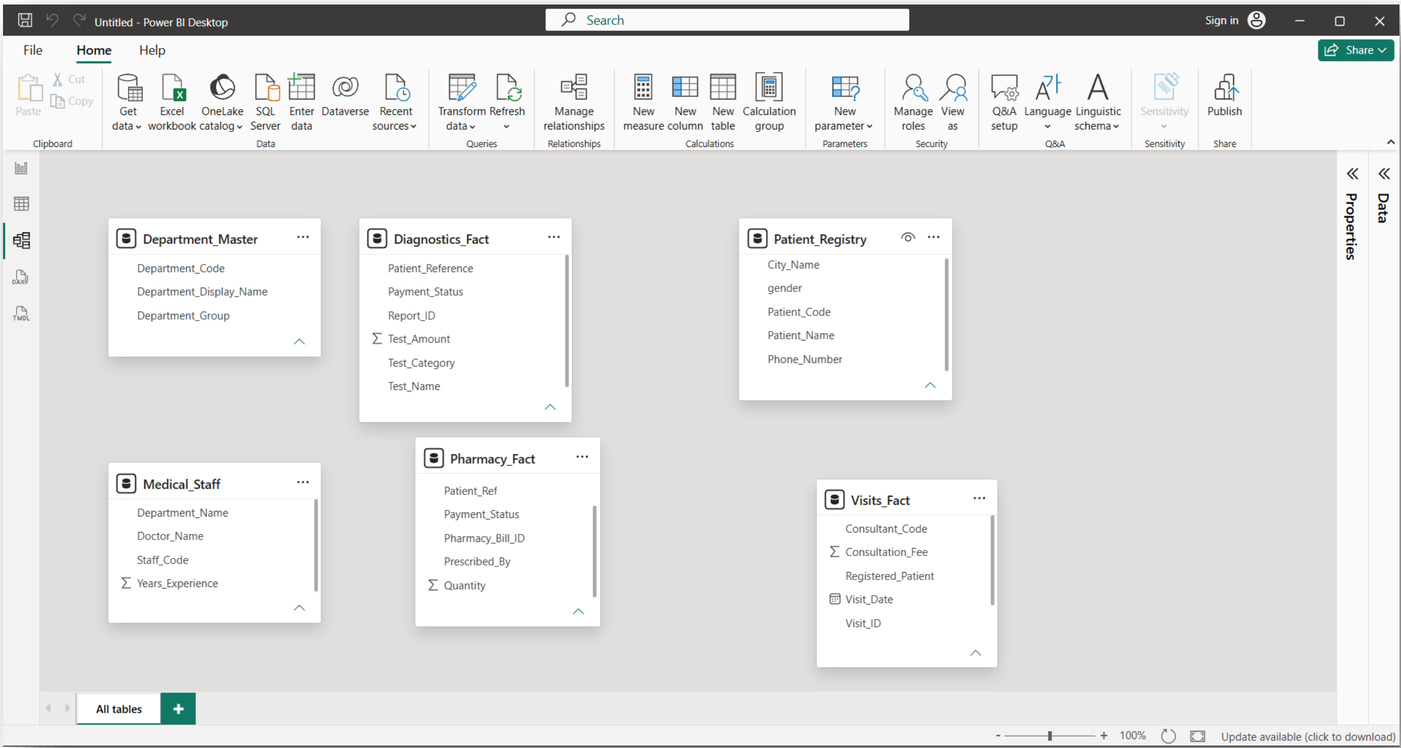
Patient_Registry[Patient_Code]

Purpose:

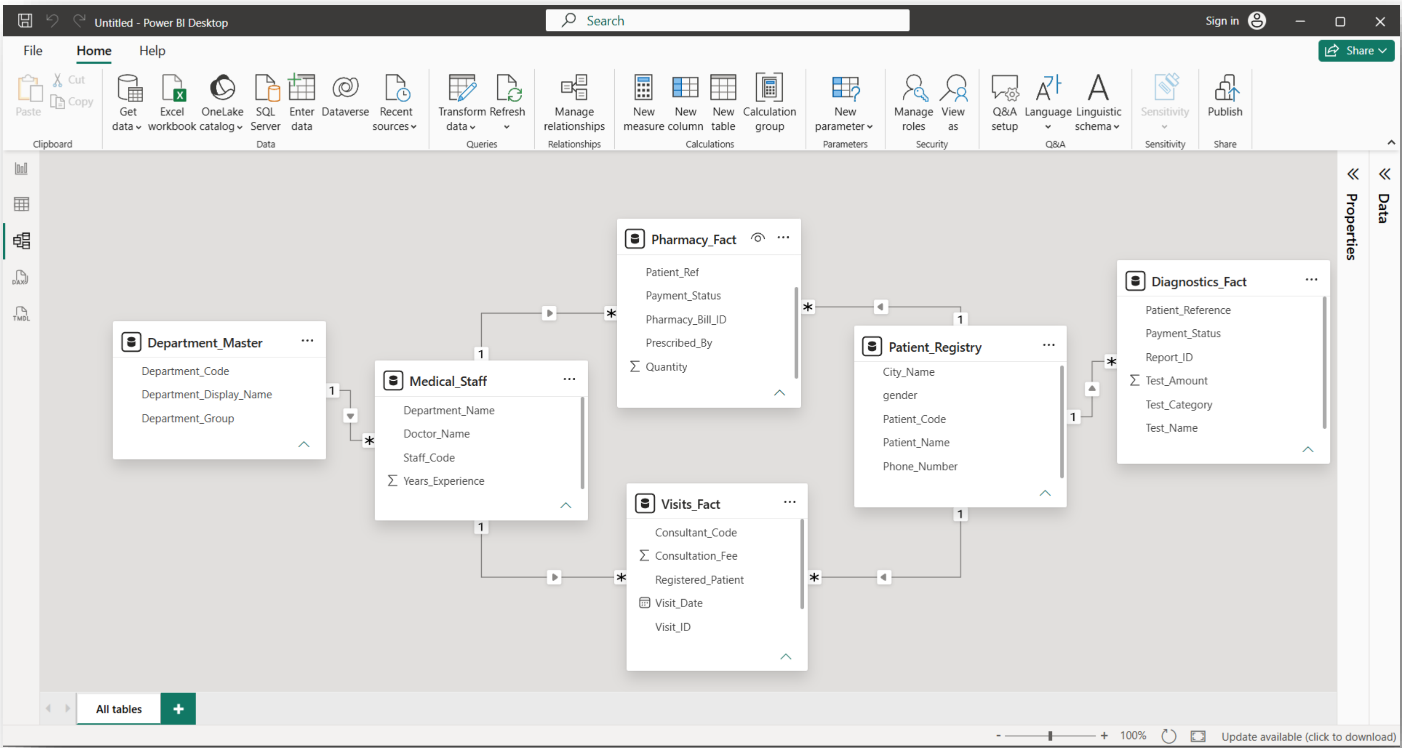
One patient can visit the hospital multiple times, creating multiple records in the Visits Fact table.

Relationship Type: Many-to-One (*:1)

Before Modeling and Realationships with each other Table Model View Shows :



After Modeling and create Realationships With each other Model View Shows :



Task 4: DAX Calculated Columns & Measures

What is DAX?

DAX (Data Analysis Expressions) is a formula language used in Power BI to create:

- Calculated Columns
- Measures
- Calculated Tables

1. Difference Between Calculated Column and Measure

Calculated Column

A calculated column performs a calculation for **each row** of a table.

Measure

A measure performs calculations dynamically based on filters and visuals.
The result changes when you use:

- Slicers
- Filters
- Charts

2. DAX Measures :

Measure 1: Total Amount / Total Sales

Total Sales = SUM(Orders[Sales])

Measure 2: Average Sales Amount

Average Sales Amount = AVERAGE(Orders[Sales])

Measure 3: Total Number of Records

Total Orders = COUNTA(Orders[Order ID])

Measure 4: Max Sales Value

Max Sales = MAX(Orders[Sales])

Measure 5: Min Sales Value

Min Sales Value = MIN(Orders[Sales])

Measure 6: Total Revenue

Total Revenue = sum(Orders[Total Revenue])

Measure 7 : Total Profit

Total Profit = SUM(Orders[Profit])

Measure 8 : Profit %

Profit % = DIVIDE([Total Profit], [Total Revenue Amount] , 0)

3. How to Create a Measure in Power BI

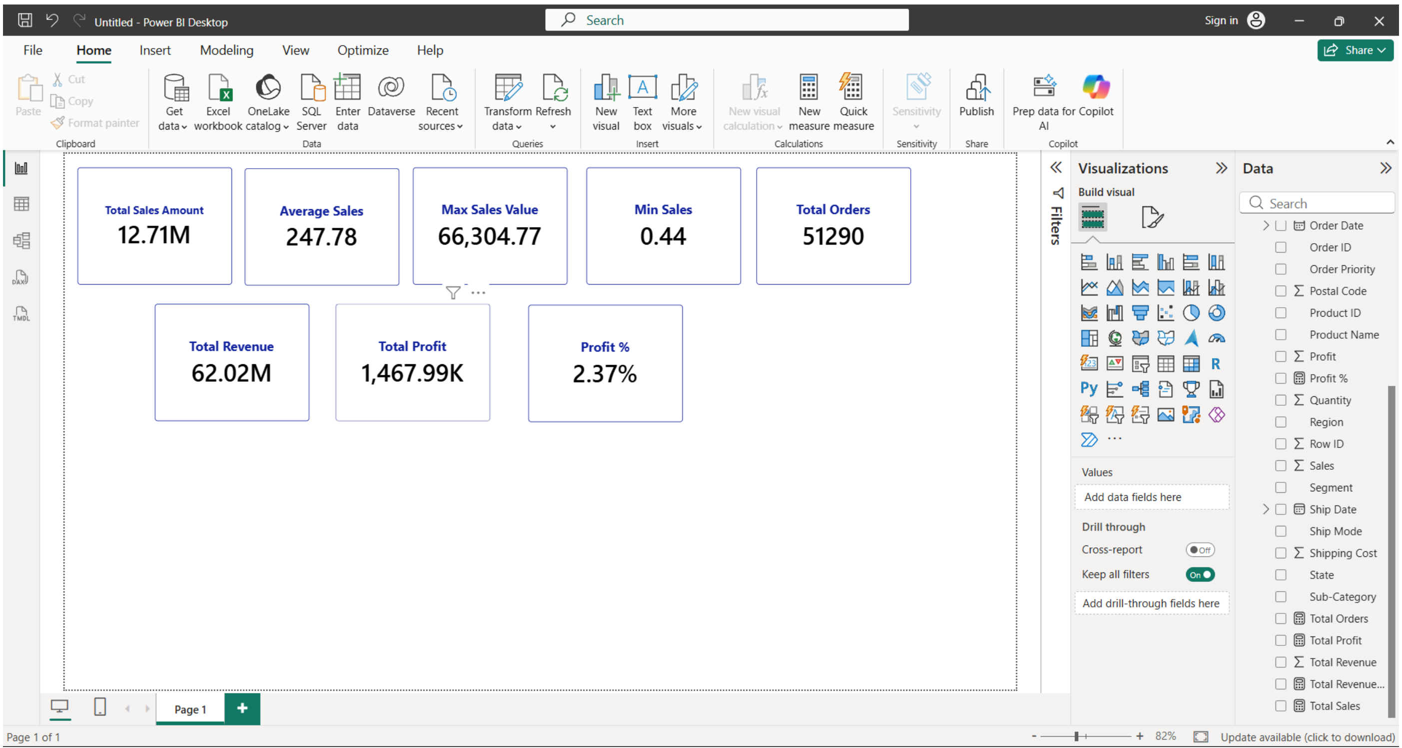
1. Go to the Data View or Report View.
2. Select the table where you want to create the measure.
3. Right-click the table.
4. Select New Measure.
5. Enter the DAX formula.
6. Press **ENTER**

4. Display the Results

To display the results:

1. Go to **Report View**.
2. Select the **Card** visual.
3. Drag the measure into the card.

Output :



Task 5: Power BI Visualizations

Objective

The objective of this task is to create different **Power BI visualizations** to represent the data clearly and identify trends, comparisons, patterns, and business insights.

What is Data Visualization?

Data Visualization is the graphical representation of data using charts, graphs, tables, and other visual elements.

Power BI visualizations help convert raw data into meaningful and easy-to-understand information.

Visualizations to Create

Based on My **Financials dataset**, I Have create the following visualizations.

1. Line Chart – Total Sales by Month

Visual:

Line Chart

Fields:

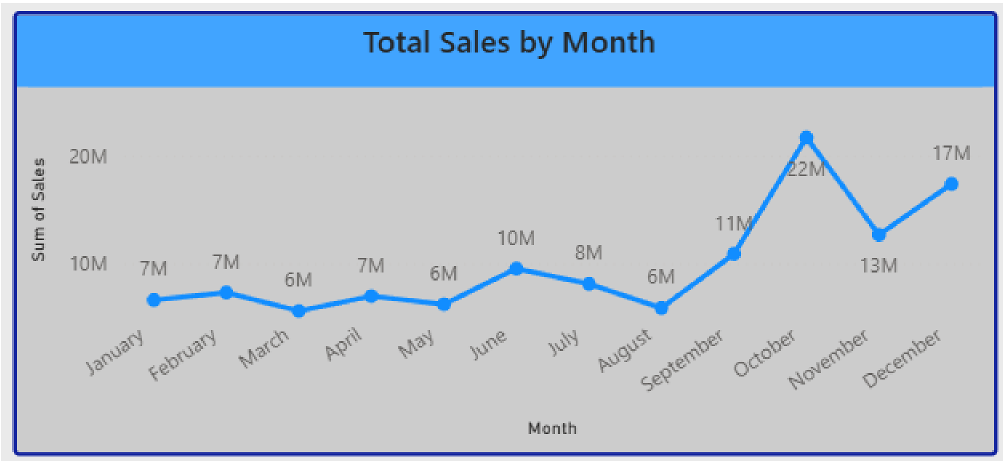
- **X-Axis:** Month Name
- **Y-Axis:** Sales

Title:

Total Sales by Month

Purpose:

This visualization shows the monthly sales trend and helps identify months with high and low sales.



2. Clustered Column Chart – Total Sales by Segment and Year

Visual:

Clustered Column Chart

Fields:

- **X-Axis: Segment**

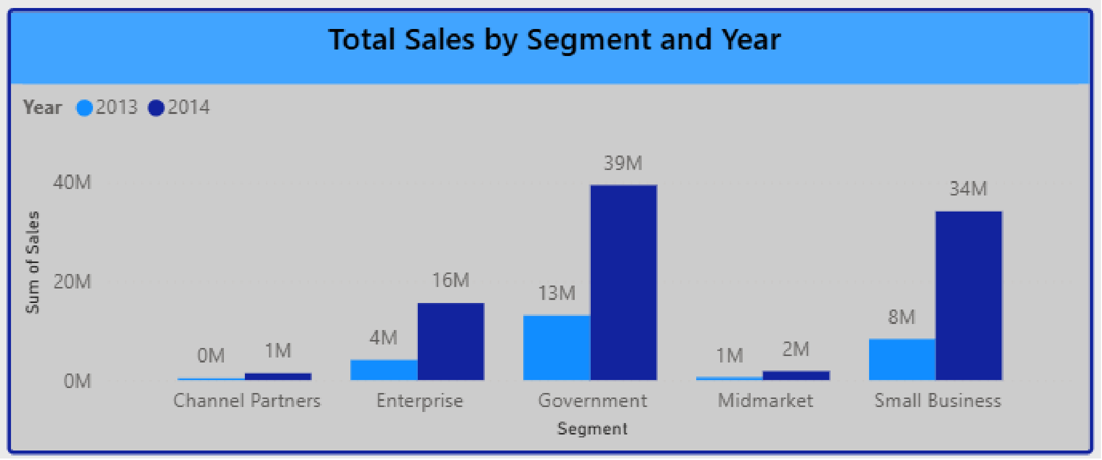
- **Y-Axis: Sales**
- **Legend: Year**

Title:

Total Sales by Segment and Year

Purpose:

This chart compares sales performance across different business segments and years.



3. Donut Chart – Profit by Product

Visual:

Donut Chart

Fields:

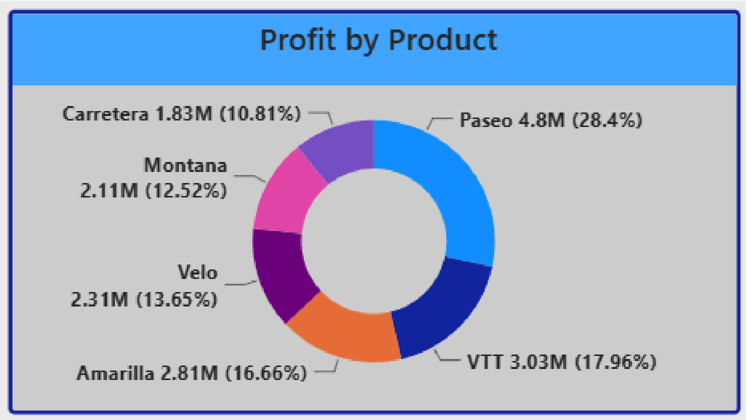
- **Legend: Product**
- **Values: Profit**

Title:

Profit by Product

Purpose:

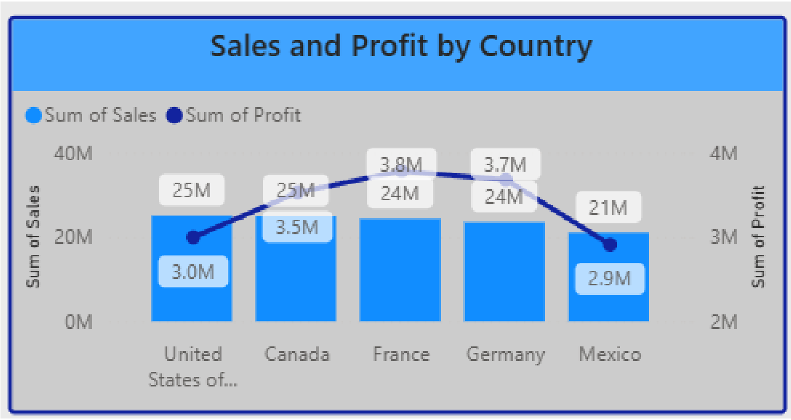
This visualization shows the contribution of each product to total profit.



4. Line and Clustered Column Chart – Sales and Profit by Country
Visual:
Line and Clustered Column Chart
Fields:

- **X-Axis: Country**
- **Column Y-Axis: Sales**
- **Line Y-Axis: Profit**

Title:
Sales and Profit by Country
Purpose:
This visualization compares sales and profit across different countries.



5. Table Visual – Product, Segment and Gross Sales
Visual:
Table
Fields:

- **Product**
- **Segment**
- **Gross Sales**

Title:
Product-wise Gross Sales Details
Purpose:
This visual provides detailed information about product sales and segments.

Product	Segment	Sum of Gross Sales
Amarilla	Small Business	50,24,400.00
Carretera	Small Business	42,75,000.00
Montana	Small Business	71,17,950.00
Paseo	Small Business	1,23,21,000.00
Velo	Small Business	71,72,250.00
VTT	Small Business	1,00,31,100.00
Amarilla	Midmarket	2,77,620.00
Carretera	Midmarket	3,64,800.00
Montana	Midmarket	3,09,540.00
Total		**12,79,31,598.50**

6. KPI Cards

1. Total Sales

1. Select the Card visual.
2. Drag the Sales column into the card.
3. Power BI will automatically calculate the Sum of Sales.

Set the title as:

Total Sales

2. Total Profit

1. Add a Card
2. Drag the Profit column.
3. Set aggregation to Sum.

Result:

Sum of Profit

Title:

Total Profit

3. Total Gross Sales

1. Add a Card.
2. Drag Gross Sales into the card.
3. Select aggregation as Sum.

Title:

Total Gross Sales

4. Total Units Sold

1. Add a Card.
2. Drag Units Sold into the card.
3. Select aggregation as Sum.

Title:

Total Units Sold

5. Profit %

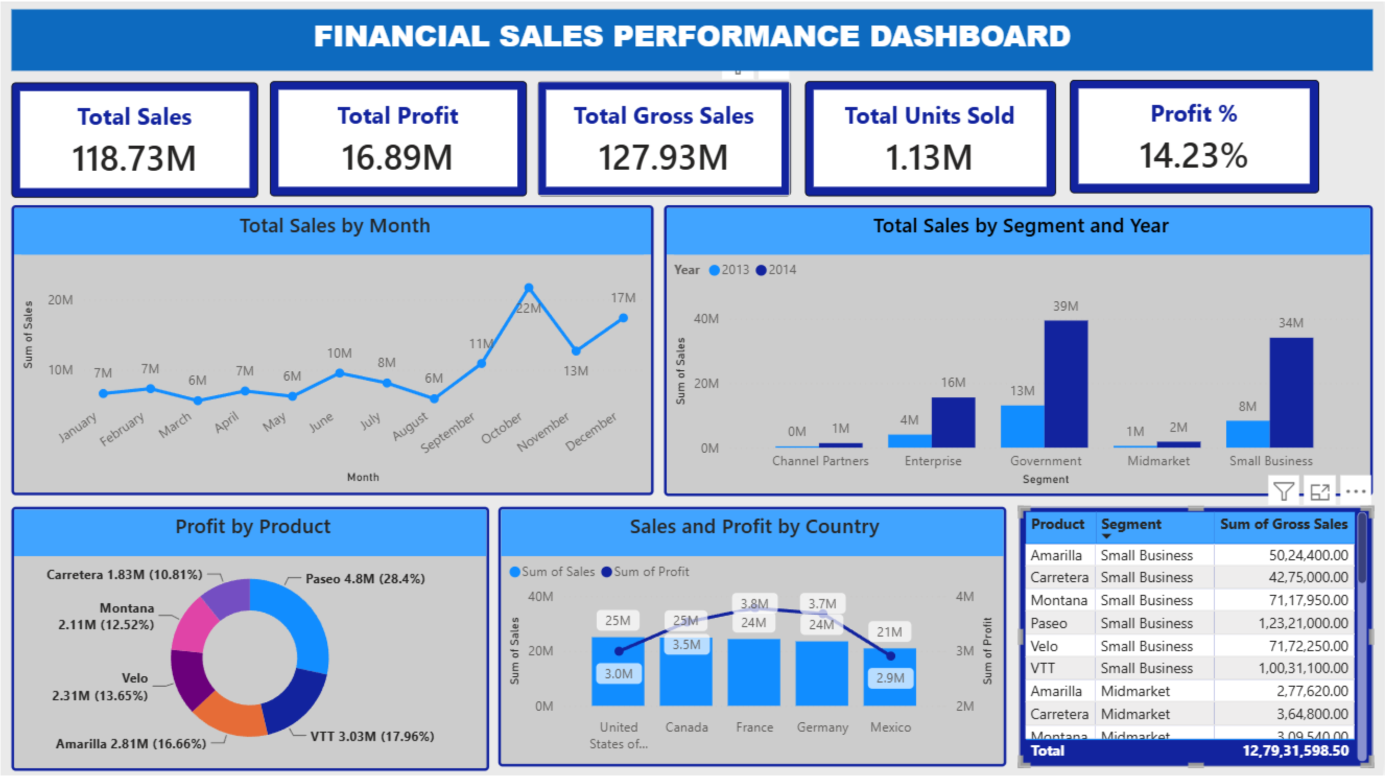
Profit % = DIVIDE([Total Profit], [Total Sales], 0)

After creating the measure:

Measure Tools → Format → Percentage

1. Add a Card.
2. Drag Measured Profit % into the card.

Final Output :



Task 6: Interactive Dashboard & Filters

Objective :

To design and develop an interactive Power BI dashboard using KPI cards, multiple visualizations, slicers, and filters to enable users to explore data, analyze business performance, identify trends, and gain meaningful insights for decision-making.

What is Visualization?

Visualization is the graphical representation of data using charts, graphs, maps, tables, and other visual elements.

In Power BI, visualizations help convert **raw data into easy-to-understand information and insights**.

Different Types of Visualizations in Power BI

1. Bar Chart

Used to compare values between different categories.

Best for:

- Comparing categories
- Ranking products
- Comparing regions

2. Column Chart

Similar to a bar chart, but displayed vertically.

Best for:

- Comparing sales
- Comparing profit
- Category-wise analysis

3. Line Chart

Used to show trends over time.

Best for:

- Monthly trends
- Yearly trends
- Revenue growth

4. Pie Chart

Shows how each category contributes to the total.

Best for:

- Percentage contribution
- Small number of categories

5. Donut Chart

Similar to a Pie Chart but has a hole in the center.

Best for:

- Contribution analysis
- Product-wise profit
- Category distribution

6. Area Chart

Shows trends and emphasizes the magnitude of values over time.

Best for:

- Sales trends
- Revenue trends
- Growth over time

7. Clustered Column Chart

Compares multiple values across categories.

Best for:

- Comparing multiple measures
- Year-wise comparison
- Category comparison

8. Stacked Column Chart
Shows the total and the contribution of different categories.
Best for:

- Contribution to total
- Category comparison

9. Combo Chart
Combines two charts, usually:

- Column Chart
- Line Chart

Best for:

- Comparing two different measures
- Comparing Sales vs Profit

10. Scatter Chart
Shows the relationship between two numerical variables.
Best for:

- Correlation analysis
- Finding outliers
- Relationship analysis

11. Treemap
Displays data using rectangles of different sizes.
Best for:

- Hierarchical data
- Category contribution
- Comparing proportions

12. Map
Displays geographical data.
Best for:

- Country-wise analysis
- State-wise sales
- Regional performance

13. Table
Displays detailed data in rows and columns.

Best for:

- Detailed information
- Record-level analysis

14. Matrix

Similar to a Pivot Table in Excel.

Best for:

- Hierarchical analysis
- Summary reports
- Comparing multiple dimensions

15. Card

Displays a single important KPI.

Best for:

- KPIs
- Important summary values

17. Slicer

A slicer is an interactive filtering visualization.

When a user selects a value, all connected visuals can update.

I have create a interactive *AMAZON PRIME SALES DASHBOARD*.

Dashboard Title

SALES ANALYSIS | AMAZON PRIME

1. KPI Cards

The dashboard contains **3 KPI Cards**.

1. Total Sales

Total Sales = sum('Amazon Prime Data'[Price(Dollar)])

After creating the measure:

1. Add a Card.
2. Drag Measured Total Sales into the card.

2. **Total Product Description**

Total Product Description = DISTINCTCOUNT('Amazon Prime Data'[Product Description])

After creating the measure:

1. Add a Card.
2. Drag Measured Total Product Description into the card.

3. Total No of Reviews

Total No of Reviews = SUM('Amazon Prime Data'[Number of reviews])

After creating the measure:

1. Add a Card.

2. Drag Measured Total no of reviews into the card.

These KPI cards provide a quick summary of the overall dataset.

2. Slicers / Filters

The dashboard contains **3 interactive slicers** on the left side.

Quality Filter

Users can filter the dashboard based on quality, such as:

- Prime Video
- Blu-ray
- DVD
- 4K
- Multi-Format

Director Filter

Users can select a specific **Director** to analyze movies and sales associated with that director.

Movie Type Filter

Users can filter the data based on different **Movie Types**.

These filters allow all connected visuals and KPI cards to update dynamically.

3. Visualizations Used

The dashboard contains more than the required **4 visualizations**.

Visualization 1: Line Chart

Title:

Total Movie Count by Release Year (2010–2023)

This chart displays the number of movies released each year.

Purpose:

- Identify movie release trends.
- Find years with the highest number of movies.
- Compare movie counts over time.

Visualization 2: Bar Chart

Title:

Total Sales by Movie Type

This visualization compares total sales across different movie types.

Purpose:

- Identify the highest-selling movie type.
- Compare sales performance between movie classifications.

Visualization 3: Donut Chart

Title:

Total Sales by Quality

This chart shows how total sales are distributed among different movie quality formats.

Purpose:

- Analyze the contribution of each quality type.
- Identify the quality category generating the highest sales.

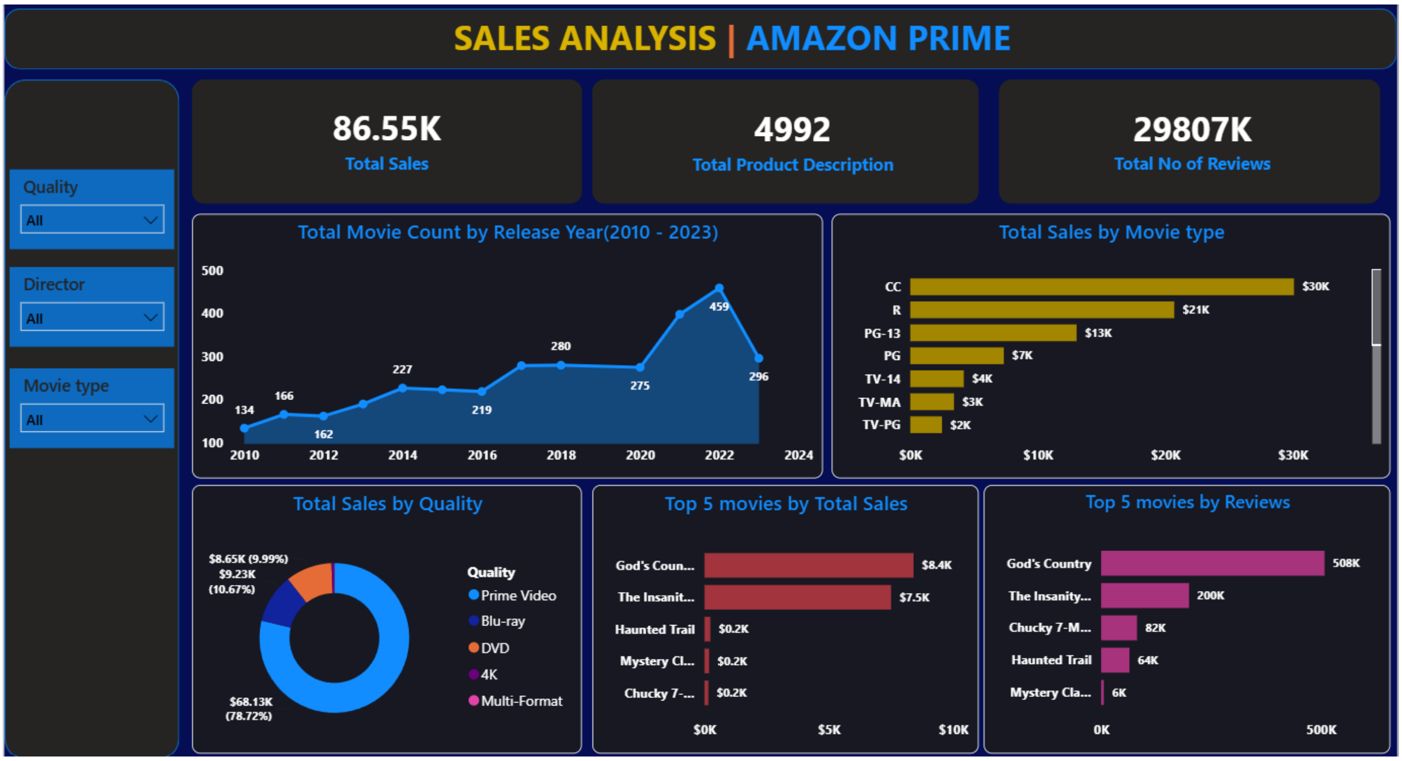
Visualization 4: Bar Chart
Title:
Top 5 Movies by Total Sales
This visualization displays the five movies generating the highest sales.
Purpose:

- Identify the best-selling movies.
- Compare the sales performance of the top movies.

Visualization 5: Bar Chart
Title:
Top 5 Movies by Reviews
This chart displays the five movies with the highest number of reviews.
Purpose:

- Identify the most reviewed movies.
- Analyze audience engagement and popularity.

Final Output of Dashboard :



Task 7: Business Insights & Recommendations

Objective :

The objective of this task is to analyze the Power BI dashboard, identify meaningful business insights, recognize important trends and patterns, and provide data-driven recommendations.

Business Insights from the Amazon Prime Dashboard

Insight 1: Movie Count Increased Significantly Over Time

The **Total Movie Count by Release Year** chart shows an overall increasing trend in movie releases.

Insight 2: CC Movie Type Generates the Highest Sales

According to the **Total Sales by Movie Type** chart:

- **CC** generated the highest sales of approximately **$30K**.
- **R** generated approximately **$21K**.
- **PG-13** generated approximately **$13K**.
- **TV-PG** generated the lowest sales of approximately **$2K**.

Highest Performing Movie Type

🥇 **CC – approximately $30K**

Lowest Performing Movie Type

📉 **TV-PG – approximately $2K**

Insight 3: Prime Video Quality Dominates Total Sales

The **Total Sales by Quality** chart shows that:

- **Prime Video** generated approximately **$68.13K**, representing **78.72% of total sales**.
- Other quality formats such as Blu-ray and DVD contribute significantly less.

Business Recommendations

Recommendation 1: Focus More on High-Performing Movie Types

The **CC movie type generates approximately $30K**, making it the highest-performing category.

Recommendation

Amazon Prime should analyze the characteristics of successful CC movies and consider acquiring, promoting, or producing more content with similar characteristics.

Recommendation 2: Continue Investing in Prime Video Content

Prime Video quality contributes approximately:

$68.13K or 78.72% of total sales

Recommendation

The business should continue investing in Prime Video content because it is the dominant source of sales and has the strongest customer preference.

Recommendation 3: Promote Highly Reviewed Movies

God's Country has:

- **$8.4K in sales**
- **508K reviews**

Recommendation

Movies receiving high customer engagement and reviews should receive additional promotion because they have the potential to generate higher sales.